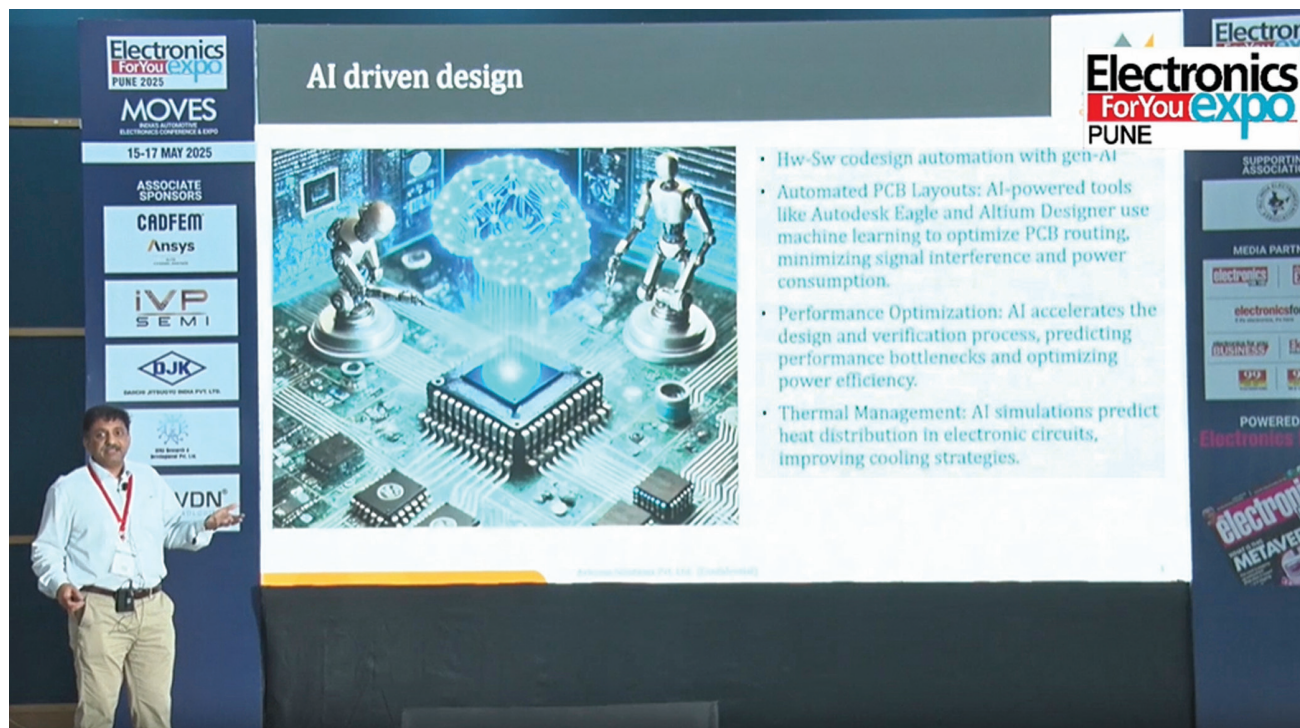


How AI Is Revolutionising EMBEDDED SYSTEMS' DESIGN

From flowcharts to firmware, the embedded design pipeline is evolving. This article examines AI's growing influence on embedded design workflows.



Tapan Kapoor, Co-Founder, Aviconn Solutions Pvt Ltd during his presentation at EFY Expo in Pune

My focus is on some generative AI experiments we conducted in our in-house R&D. Seeing very promising results, I felt it was worthwhile to share them here with a relevant audience that could potentially use this technology. With my experience, I hope to spark interest and help someone take these ideas forward. Let's dive in, as most of us are familiar with AI already.

The fusion of AI and embedded designs

A typical embedded system design begins with defining requirements and the intended application. Requirements

may exist as text or remain conceptual, forming the starting point. From there, the system is broken down into hardware and software needs.

The designer's experience plays a key role in selecting the microcontroller, peripherals, power constraints, and timing requirements. These decisions ultimately shape the hardware and software components of the system.

This is where generative AI can help. In other design spaces, complete executable models in C code are already being created directly from requirement descriptions. The system can then be implemented, executed on the target microcontroller, and analysed for bottlenecks. This